

12.536

EE25BTECH11002 - Achat Parth Kalpesh

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Question

The rank of the matrix $\begin{pmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & 2 & 2 \\ 4 & 1 & 3 & 1 \end{pmatrix}$ is

1 1

2 2

3 3

4 4

Solution

We perform row operations to reduce it to row echelon form:

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & 2 & 2 \\ 4 & 1 & 3 & 1 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 4 & 1 & 3 & 1 \end{pmatrix} \quad (1)$$

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 4 & 1 & 3 & 1 \end{pmatrix} \xleftrightarrow{R_3 \leftarrow R_3 - 4R_1} \begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 0 & -3 & 3 & 9 \end{pmatrix} \quad (2)$$

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 0 & -3 & 3 & 9 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow -\frac{1}{2}R_2} \begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & 1 & -1 & -3 \\ 0 & -3 & 3 & 9 \end{pmatrix} \quad (3)$$

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & 1 & -1 & -3 \\ 0 & -3 & 3 & 9 \end{pmatrix} \xleftrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -3 \\ 0 & -3 & 3 & 9 \end{pmatrix} \quad (4)$$

Solution

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -3 \\ 0 & -3 & 3 & 9 \end{pmatrix} \xleftrightarrow{R_3 \leftarrow R_3 + 3R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (5)$$

Now the matrix is in reduced row echelon form

The number of non-zero rows = 2

$$\text{rank}(A) = 2 \quad (6)$$